

Aerial-Ground Robots Collaborative 3D Mapping in GNSS-Denied Environments

Yufeng Yue^{1*}, Chunyang Zhao², Yuanzhe Wang², Yi Yang¹, Danwei Wang², *Fellow, IEEE*

Abstract—Collaborative heterogeneous robots are expected to perform comprehensive perception, mapping and coordination in search and rescue scenarios. The challenge of collaboration between heterogeneous robots lies in their huge differences in perception, mobility and processing capabilities. In this paper, a novel collaborative UAV-UGV mapping framework is proposed in GNSS-denied and unknown environments. The key novelty of this work is the proposing of a unified framework to formulate the UAV-UGV collaborative mapping problem with a continuous-discrete model, as well as its realization in real robotic systems. In order to project continuous space into discrete space, a novel information gain trigger scheme is proposed. The continuous space allows each robot to perform high frequency local map estimation, while discrete space describes the problem of multi-resolution hybrid map fusion. Considering the nature of data heterogeneity, a flexible probabilistic fusion algorithm is proposed that addresses the multi-resolution hybrid map fusion problem, where the local maps generated by UAV and UGV are fused based on Bayesian rule. The proposed UAV-UGV hybrid system is validated in various challenging scenarios, demonstrating its accuracy and utility in practical tasks.

I. INTRODUCTION

Compared to homogeneous robots that have identical functions, diverse capabilities of heterogeneous robots are brought together to achieve enhanced performance in search and rescue missions [1], [2]. Many current systems require partial environment information to be known as a prior, which is utilized to perform missions such as task planning [3], relative localization [4], target docking [5]. However, search and rescue applications require the robots to operate in situations without prior environmental models. In such cases, collaborative perception and mapping are critical capabilities required by multi-robot systems to understand the environment. Therefore, the premise of achieving online operation is that heterogeneous robots need to understand the scene, share their observations between UAV and UGV, and generate a globally consistent map in real time. This article provides collaborative hybrid UAV-UGV mapping as a systematic solution for heterogeneous robots perception and mapping.

This work is supported by the National Natural Science Foundation of China under Grant 62003039, 61973034, U193203, 61903034 and the CAST program under Grant No. YESS20200126. (Corresponding Author: Yufeng Yue)

¹Yufeng Yue, Yi Yang are with School of Automation, Beijing Institute of Technology, Beijing, China (email: yueyufeng@bit.edu.cn)

²Chunyang Zhao, Yuanzhe Wang, Danwei Wang are with School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore

Employing heterogeneous robots can complement their strengths, constructing a comprehensive and accurate description of the environment. In order to operate in Global Navigation Satellite System (GNSS) denied and unknown environments, each robot needs to estimate its pose and local map in high frequency [6]. On this basis, UAV and UGV share their local maps and generate a global fused map [7]. Most approaches generate the global environmental representation off-line in a centralized setting [7]–[10]. However, it is more robust for each robot to maintain and update a global map distributively, which supports online path planning and navigation from a holistic view. Considering the limited computational power on-board, especially for the UAV, higher-level objects such as collaborative mapping can be performed periodically in discrete space. In order to balance the contradiction between map fusion and limited computing power, the first challenge is to carefully design a hierarchical framework that jointly considers the tasks of local mapping and global map fusion, which should be computationally efficient and adaptable to environment.

Compared with homogeneous robots, the installation of heterogeneous sensors on UAV and UGV brings another layer of challenge. In view of high payload and power supply, UGV is equipped with a high-precision 3D Lidar to capture 360-degree environmental information. However, UGV can easily fall into a blind spot or be blocked by obstacles, leading to failure of mapping missions. In contrast, UAV has high flexibility and maneuverability, but only has limited payload. This inherent feature makes the UAV can only be equipped with light low-resolution sensors. Due to the huge difference in sensing resolution, detection range, and field of view (FoV), map fusion between hybrid robots is a challenging problem. Up to date, most existing approaches on UAV-UGV mapping [11]–[13] assume that UAV and UGV are equipped with same type of sensors and only need to fuse homogeneous data. Considering the nature of data heterogeneity, the second challenge is to propose a flexible probabilistic fusion algorithm that integrates and combines the multi-resolution local maps from hybrid robots.

Above limitations hinge on the individual capabilities of the robots, either in distributed information sharing strategy, or multi-resolution hybrid map fusion. This interwoven nature of the problem motivates our research towards a systematic collaborative UAV-UGV mapping approach. The main contributions are listed as follows:

- A collaborative hybrid UAV-UGV mapping framework is proposed based on continuous-discrete model. The continuous space achieves real time local map generation, while the discrete cycle performs sporadically map

fusion based on information gain.

- A flexible probabilistic fusion algorithm is proposed to address the multi-resolution hybrid map fusion problem, where the local maps generated by UAV and UGV are fused based on Bayesian rule.
- The proposed UAV-UGV hybrid system is validated in various experimental scenarios, demonstrating its accuracy and utility in practical tasks.

The rest of the paper is organized as follows. Section II reviews the existing approaches. Section III defines the problem. Section IV explains the theoretical basis for collaborative UAV-UGV mapping. Section V shows the experimental results.

II. RELATED WORK

In this section, an extensive survey of existing approaches related to heterogeneous robots perception and mapping is provided.

A. Homogeneous robots mapping

Reference [14] summarizes different solutions for multiple UGVs mapping. In challenging 3D scenarios, submap based map fusion algorithm is proposed in [15]. Then, the authors in [16] propose map updating algorithm based on Octree structure. In the unstructured environment, a probabilistic fusion strategy is proposed for heterogeneous map matching in [17]. In day and night environment, [18] has proposed a collaborative mapping framework based on multi-modal environment perception. To provide the view of eye in the sky, there are also approaches that utilizes multiple UAVs for environmental mapping. An introduction of multiple UAVs environmental mapping is reviewed in [19]. Reference [20] develops a system consisting of multiple drones that can perform autonomous navigation and 3D mapping in GNSS-denied environments. Reference [21] proposes a framework to implement vision-based multi-UAV collaborative mapping in simulation. In [10], multi-UAV cooperative SLAM is proposed, but the map fusion is handled offline in the central station.

These approaches made significant progress in collaborative mapping for multiple UAVs or UGVs. However, they only focus on the cooperation between same type of robot platforms, and the advantages of UAV and UGV have not been combined. In addition, existing references have not considered the problem of designing a systematic framework of generating local maps and global map with continuous-discrete model.

B. Heterogeneous Robots Mapping

The work of UAV-UGV teaming is pioneered by [22], where maps generated by aerial robots are used to design navigation controllers and plan missions for the UGV. A detailed evaluation of various relative localization algorithms for UAV-UGV can be found in [9]. To leverage the flexible perception ability of UAV, reference [11] provides bird eye

view for UGV. A cooperative surveillance system is proposed that allows the UAV to land on the UGV [23].

In order to achieve detailed environmental mapping, ground and aerial robots are utilized to perform collaborative mapping in an earthquake-damaged building [24]. Similarly, UAV is located in the coordinate frame of UGV and a global map is generated in [8]. In digital construction, the point cloud from UAV and UGV are registered and stitched using 2D image features [7]. To realize precision farming, a aerial-ground collaborative 3D mapping framework is proposed in [12]. More recently, UAV-UGV exploration [25] is proposed that aims to solve the area coverage and navigation problem.

To make a short summary, aforementioned collaborative mapping algorithms for UAV-UGV team perform offline data processing and 3D reconstruction. Furthermore, they have not considered the fusion between multi-resolution maps.

III. SYSTEM FRAMEWORK

A. System Architecture

The objective of this work is to develop a collaborative hybrid UAV-UGV mapping framework that enables hybrid robots to generate a consistent global map. The overview of the system architecture consists of two levels: *single robot continuous mapping level* and *UAV-UGV discrete map fusion level*. Since the hybrid systems operate in a distributed configuration, the organization of this paper follows a hierarchical structure from the single robot level to the collaborative level.

B. Continuous-Discrete Space Definition

1) *Single Robot Continuous Mapping Model*: The continuous mission involves single robot localization and mapping, which has been well studied and many efficient approaches have been developed. Although UAV and UGV are equipped with different sensors, the mapping process shares the same theory and can be formulated in (1).

Problem 2: For each robot r , the objective is to estimate its local map $\mathbf{M}_r^{[t_0, t_n]}$, trajectory $\mathbf{X}_r^{[t_0, t_n]}$ given the sensor observations $\mathbf{Z}_r^{[t_0, t_n]}$.

$$p(\mathbf{M}_r^{[t_0, t_n]}, \mathbf{X}_r^{[t_0, t_n]} | \mathbf{Z}_r^{[t_0, t_n]}) = \prod_{k=1}^{N_f} p(\mathbf{M}_r^{[t_{k-1}, t_k]}, \mathbf{X}_r^{[t_{k-1}, t_k]} | \mathbf{Z}_r^{[t_{k-1}, t_k]}) \quad (1)$$

Given local observations $\mathbf{Z}_r^{[t_0, t_n]}$, each robot r estimates its local map $\mathbf{M}_r^{[t_0, t_n]}$ and poses $\mathbf{X}_r^{[t_0, t_n]}$ in continuous space $[t_0, t_n]$. Single robot mapping process is factorized into a product of mapping process in a series of time intervals in (1), where $[t_0, t_n] = \sum_{k=1}^{N_f} [t_{k-1}, t_k]$. Fig. 1 denotes the process of single robot local mapping (in red), which is detailed in Sec. IV-A.

2) *UAV-UGV Discrete Map Fusion*: For collaborative mapping missions, the local maps are shared in discrete manner to increase the efficiency and robustness. Robots only exchange the generated local maps to reduce the amount of shared data, thereby reducing communication consumption.

Problem 3 Given heterogeneous robots in GNSS-denied and unstructured environments, the input is the local map $\mathbf{M}_r^{[t_0, t_n]}$, and its neighboring robot's map $\mathbf{M}_s^{[t_0, t_n]}$ generated in

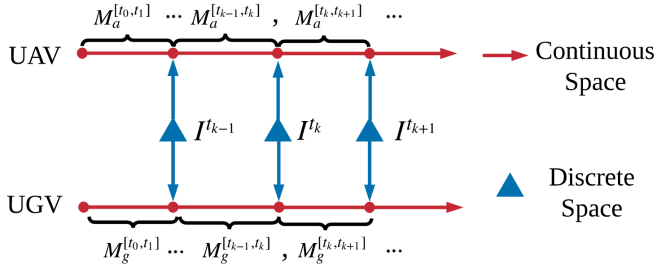


Fig. 1. The continuous-discrete model for heterogeneous robots mapping.

continuous time domain $[t_0, t_n]$, the objective of robot r is to decide a set of trigger point $\mathbf{I}_r^{t_0:t_n}$ to generate the global map $\mathbb{M}_r^{t_0:t_n}$ at discrete time slots $t_0 : t_n$.

$$p(\mathbb{M}_r^{t_0:t_n} | \mathbf{I}_r^{t_0:t_n} | \mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]}) \quad (2)$$

In (2), local maps $\mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]}$ are generated in continuous time domain $[t_0, t_n]$ with consecutive incoming raw sensor data. Under limited bandwidth and computational power, local maps are shared and fused in a set of discrete time slots $t_0 : t_n = \{t_0, \dots, t_k, \dots, t_n\}$. To increase robustness, each robot r generates its own global map $\mathbb{M}_r^{t_0:t_n}$. Its neighboring robot s also maintains a global map $\mathbb{M}_s^{t_0:t_n}$. In the following content, we focus on robot r , which can be UGV or UAV. Then, the paper calculates a set of trigger points $\mathbf{I}_r^{t_0:t_n}$ to generate global map $\mathbb{M}_r^{t_0:t_n}$. Fig. 1 denotes the process of UAV-UGV discrete mapping (in blue).

To estimate $\mathbf{I}_r^{t_0:t_k}$ and $\mathbb{M}_r^{t_0:t_k}$, (2) is further factorized into two subproblems, *Information Gain Triggered Map Sharing* $p(\mathbf{I}_r^{t_0:t_k} | \mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]})$, and *Multi-Resolution Hybrid Map Fusion* $p(\mathbb{M}_r^{t_0:t_k} | \mathbf{M}_r^{t_0:t_n}, \mathbf{I}_r^{t_0:t_n}, \Phi(\mathbf{M}_s^{t_0:t_n}))$.

$$p(\mathbb{M}_r^{t_0:t_n} | \mathbf{I}_r^{t_0:t_n} | \mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]}) = \underbrace{p(\mathbf{I}_r^{t_0:t_k} | \mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]})}_{\text{Information gain triggered map sharing}} \cdot \underbrace{p(\mathbb{M}_r^{t_0:t_k} | \mathbf{M}_r^{t_0:t_n}, \mathbf{I}_r^{t_0:t_n}, \Phi(\mathbf{M}_s^{t_0:t_n}))}_{\text{Multi-resolution hybrid map fusion}} \quad (3)$$

Information Gain Triggered Map Sharing: To project continuous space $\mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]}$ to discrete domain $\mathbf{I}_r^{t_0:t_n}$, the information gain triggered approach is introduced. A trigger slot $\mathbf{I}_r^k$ is decided based on the information gain during a time interval. When a certain threshold is satisfied at t_k , the map fusion will be triggered. As formulated in (4), $\mathbf{I}_r^k$ is the k -th trigger point for map fusion, which is decided by the information gain $[t_{k-1}, t_k]$. The details is given in Sec. IV-B.

$$p(\mathbf{I}_r^{t_0:t_n} | \mathbf{M}_r^{[t_0, t_n]}, \mathbf{M}_s^{[t_0, t_n]}) = \prod_{k=1}^{N_I} p(\mathbf{I}_r^{t_k} | \mathbf{M}_r^{[t_{k-1}, t_k]}, \mathbf{M}_s^{[t_{k-1}, t_k]}) \quad (4)$$

Multi-Resolution Hybrid Map Fusion: The process of combining multi-resolution local maps to generate the global map is formulated in (5). The overall process can be decom-

posed into the product of serial map fusion at each trigger point t_k , which is detailed in Sec. IV-C.

$$p(\mathbb{M}_r^{t_0:t_n} | \mathbf{M}_r^{t_0:t_n}, \mathbf{I}_r^{t_0:t_n}, \Phi(\mathbf{M}_s^{t_0:t_n})) = \prod_{k=1}^{N_I} p(\mathbb{M}_r^{t_k} | \mathbf{M}_r^{t_k}, \mathbf{I}_r^k, \Phi(\mathbf{M}_s^{t_k})) \quad (5)$$

IV. UAV-UGV COLLABORATIVE MAPPING

A. Single Robot Continuous Mapping

In this subsection, each robot incrementally builds a local volumetric occupancy map when moving around the environment. To represent the confidence and richness of the map, overall map volumetric information (MVI) is defined and calculated.

1) *Occupancy Grid Mapping:* Given the input of 3D point cloud, UAV and UGV can generate the same type of volumetric map $\mathbf{M}$. The occupancy map $\mathbf{M}$ is composed of a set of voxels $\mathbf{M}^n$, where $n \in \{1, N\}$ and N is number of all voxels. $p(\mathbf{M}^n | \mathbf{Z}_{1:t})$ is the probability update model, and the value is within the range of $[0, 1]$. The updating of (6) depends on current observation model $p(\mathbf{M}^n | \mathbf{Z}_t)$, previous estimation model $p(\mathbf{M}^n | \mathbf{Z}_{1:t-1})$ and prior probability model $p(\mathbf{M}_0^n)$. $p(\mathbf{M}_0^n)$ denotes the initial occupancy probability of each voxel and is set as 0.5. Then, occupancy probability model $p(\mathbf{M}^n | \mathbf{Z}_{1:t})$ is updated in (6).

$$p(\mathbf{M}^n | \mathbf{Z}_{1:t}) = [1 + \frac{1 - p(\mathbf{M}^n | \mathbf{Z}_t)}{p(\mathbf{M}^n | \mathbf{Z}_t)} \frac{1 - p(\mathbf{M}^n | \mathbf{Z}_{1:t-1})}{p(\mathbf{M}^n | \mathbf{Z}_{1:t-1})} \frac{p(\mathbf{M}_0^n)}{1 - p(\mathbf{M}_0^n)}]^{-1} \quad (6)$$

2) *Map Volumetric Information:* To measure the uncertainty of occupancy grid map, information entropy is introduced, which is a basic concept in information theory to quantify the average level of uncertainty. In this paper, the probability value $p(\mathbf{M}^n)$ is considered as a random variable (RV), and $H(\mathbf{M}^n)$ is defined to represent the information entropy of the RV in (7).

$$H(\mathbf{M}^n) = -p(\mathbf{M}^n) \log_2 p(\mathbf{M}^n) - \bar{p}(\mathbf{M}^n) \log_2 \bar{p}(\mathbf{M}^n) \quad (7)$$

As entropy measures the uncertainty of a voxel, we define the volumetric information (VI) as the confidence level of probabilistic information enclosed in a voxel. To denote the volumetric information, we introduce a monotonic increasing function $\mathbf{V}(\mathbf{M}^n)$ in (8).

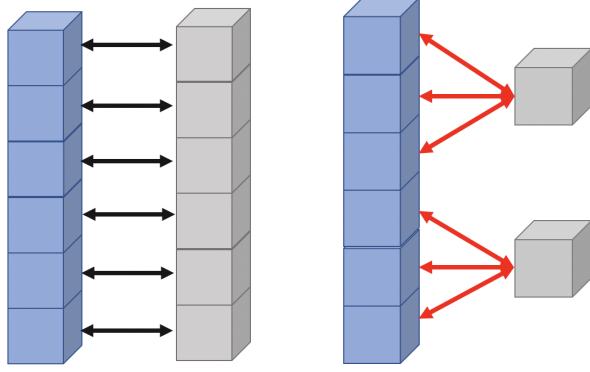
$$\mathbf{V}(\mathbf{M}^n) = 1 - H(\mathbf{M}^n), \quad H(\mathbf{M}^n) \in (0, 1] \quad (8)$$

In occupancy grid map, all the voxels are independent and probability updating process is performed independently. Therefore, the overall map volumetric information (MVI) is defined as the summation of volumetric information of each voxel.

$$\mathbf{V}(\mathbf{M}) = \sum_{n=1}^N \mathbf{V}(\mathbf{M}^n), \quad \mathbf{V}(\mathbf{M}^n) \in [0, 1] \quad (9)$$

B. Information Gain Triggered Map Sharing

we present the process of deriving information gain triggered map sharing algorithm. The map fusion trigger points $\mathbf{I}_r^{t_0:t_n}$ defined in (4) is the set of time slots to trigger robots to broadcast local maps and perform map fusion.



(a) Hard one-to-one fusion strategy (b) Flexible multi-resolution fusion

Fig. 2. The difference between hard one-to-one fusion (in black) and flexible multi-resolution fusion (in red).

1) *Map Information Gain*: Here, we introduce the concept of map information gain (MIG), which is the difference between map volumetric information (MVI) at two time slots. As defined in (10), map information gain is measured by the newly obtained information within a time interval, which will be used as a metric to calculate the trigger point. At time t , the map volumetric information of $\mathbf{M}$ is denoted as $\mathbf{V}(\mathbf{M}^t)$. Then, map information gain $\mathbf{G}(t)$ is defined as the amount of volumetric information obtained in an interval time $[t_k, t]$, where t_k denotes the previous map fusion trigger time point.

$$\mathbf{G}(t, t_k) = \alpha[\mathbf{V}(\mathbf{M}^t) - \mathbf{V}(\mathbf{M}^{t_k})] \quad (10)$$

Considering the diverse perception ability of UAV and UGV, different weights are introduced. More specifically, α_a is greater than α_g because UAV's sensor range and FOV are limited, and UAV provides environmental information that UGV cannot access.

$$\mathbf{G}_{ugv}(t|t \in [t_k, \infty)) = \alpha_g \cdot [\mathbf{V}(\mathbf{M}_g^t) - \mathbf{V}(\mathbf{M}_g^{t_k})] \quad (11)$$

$$\mathbf{G}_{uav}(t|t \in [t_k, \infty)) = \alpha_a \cdot [\mathbf{V}(\mathbf{M}_a^t) - \mathbf{V}(\mathbf{M}_a^{t_k})] \quad (12)$$

2) *Human Interface*: Since map information gain is measured by the newly obtained information, when the robot is static or moving slowly, the map fusion frequency tend to be very low. In human-robot teaming, the human teammate may need to request updating the global map at any time. Therefore, the paper introduces a factor $\mathbf{C}_o$ that allows human operator to trigger the operation. This shared control scheme allows hybrid systems to adapt to human intention, which makes the human-robot interaction and coordination possible.

3) *Trigger Map Fusion*: The map fusion process can be triggered by map information gain or human interface. The physical meaning is that the map fusion is triggered when there is sufficient newly obtained map information, or when the human teammate sends a request.

$$\mathbf{I}_{k+1}(t|t \in [t_k, \infty)) = \inf[\max(\mathbf{G}_r(t), \mathbf{C}_o(t)) > \xi] \quad (13)$$

C. Multi-Resolution Hybrid Map Fusion

Upon receiving the neighboring local maps, robot r estimates the posterior of generating a global map.

1) *Multi-Resolution Data Association*: Following (5), the problem of fusing two local maps $\mathbf{M}_r^{t_k}$ and $\mathbf{M}_s^{t_k}$ at a given trigger point $\mathbf{I}_r^{t_k}$ is simplified in (14).

$$p(\mathbb{M}_r^{t_k} | \mathbf{M}_r^{t_k}, \mathbf{I}_r^{t_k}, \Phi(\mathbf{M}_s^{t_k})) = p(\mathbb{M}_r^{t_k} | \mathbf{M}_r^{t_k}, \Phi(\mathbf{M}_s^{t_k})) \quad (14)$$

At time $t = t_0$, robot r has not received local map from robot s , global map $\mathbb{M}_r^{t_0}$ maintained by robot r equals to its local map $\mathbf{M}_r^{t_0}$. The transformation function Φ is estimated with our previous work of [17]. To simplify, the transformed local map is denoted as $\Phi(\mathbf{M}_s^{t_k}) = \mathbf{M}_s$ and current robot map is denoted as $\mathbf{M}_r$, then, (14) is rewritten as $p(\mathbb{M}_r | \mathbf{M}_r, \mathbf{M}_s)$.

To establish data association between multi-resolution maps, the multiple data association strategy is proposed. Since the voxel-wise correspondences are independent, the probability distribution of $p(\mathbb{M}_r | \mathbf{M}_r, \mathbf{M}_s)$ can be factorized into (15), where voxel $\mathbf{M}_r^i$ corresponds to a set of voxels $\mathbf{M}_s^{c_j}$.

$$p(\mathbb{M}_r | \mathbf{M}_r, \mathbf{M}_s) = \prod_{i=1} p(\mathbf{M}_r^i | \mathbf{M}_r^i, \mathbf{M}_s^{c_j}) \quad (15)$$

where $\mathbf{M}_s^{c_j}$ is consisted of a set of voxels $\mathbf{M}_s^{c_j} = \{\mathbf{M}_s^j, j \in c_j\}$. Based on total probability theory, (15) is rewritten into (16).

$$p(\mathbf{M}_r^i | \mathbf{M}_r^i, \mathbf{M}_s^{c_j}) = \sum_j^{c_j} w_{i,j} \cdot p(\mathbf{M}_r^i, e_{i,j} | \mathbf{M}_r^i, \mathbf{M}_s^j) \quad (16)$$

To represent data association between $\mathbf{M}_r^i$ and $\mathbf{M}_s^j$, a binary variable $\{e_{i,j}, j \in c_j\}$ is introduced, where $e_{i,j} = 1$ if $\mathbf{M}_r^i$ corresponds to $\mathbf{M}_s^j$ and equals to 0 otherwise. In (16), each matching pair is assigned with a weight $w_{i,j}$, which measures the importance of data association and is set as $w_{i,j} = e^{-\|\mathbf{M}_s^j - \mathbf{M}_r^i\|^2}$.

V. EXPERIMENTAL RESULTS

Real world experiments in GNSS-denied and unstructured scenarios are investigated, including NTU Rainforest (Sec. V-B), Two-Level Building (Sec. V-C).

A. Experiments Overview

1) *Experiment Setup*: Experiments are conducted by operating UAV and UGV in two different scenarios in Nanyang Technological University. All algorithms are executed on the onboard platform for single robot SLAM and collaborative mapping. For each robot, the map fusion is triggered independently, each robot updates and maintains its global map.

- **NTU Rainforest**: the fully unstructured rainforest at NTU campus, heterogeneous robots seek to map the environment.
- **Two-Level Building**: a two-level building at NTU campus, when UGV was unable to go through the staircase, UAV flew over it.

Fig. 3 presents the settings and main characteristics of each experiment.

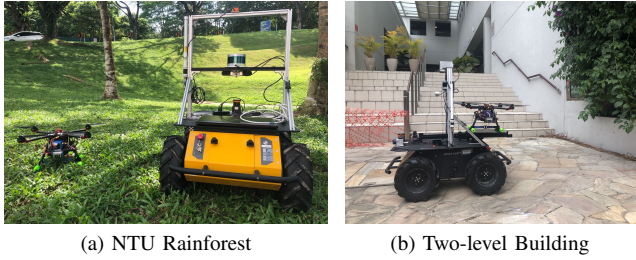


Fig. 3. Experimental environments and settings.

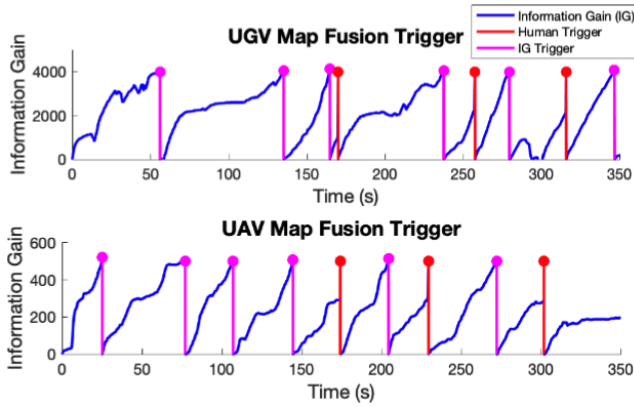


Fig. 4. The map fusion trigger slots in NTU Rainforest.

2) *Hardware Platform*: UGV platform is equipped with the Intel Core i7-6700HQ CPU @ 2.60GHz CPU with a 3D Velodyne LiDAR. UAV Platform is equipped with the Intel NUC 6i7KYK CPU @ 2.60GHz CPU with a stereo camera. The communication between robots is established by long range Wi-Fi with limited bandwidth. LOAM [26] is applied for UGV robot pose estimation. ORB SLAM [27] is applied for UAV pose estimation. Octomap [28] is utilized for 3D occupancy mapping at a resolution of 0.1m.

3) *Performance Analysis*: As no article has addressed the real time heterogeneous robots mapping problem, we first demonstrate qualitative results. The overall performance for accuracy and efficiency. For map fusion accuracy, the average entropy is compared against Jessup [16]. Since the code of [16] is not available, we implemented it based on the standard Point Cloud Library [29]. Then, the time usage on high level and low level decision cycles are summarized to show the efficiency.

B. NTU Rainforest

As shown in Fig. 5, the hybrid robots operate under a dense forest canopy, which is a GNSS-denied and unstructured 3D environment that contains trees and steep slopes.

1) *Trigger Time Line*: The information gain triggered map sharing results are summarized in Fig. 4. The blue curve represents the information gain in continuous space, purple lines represents map fusion triggered by information gain, and red lines indicates operator's command. Map fusion has been triggered 9 times in 350 seconds, including 3 times

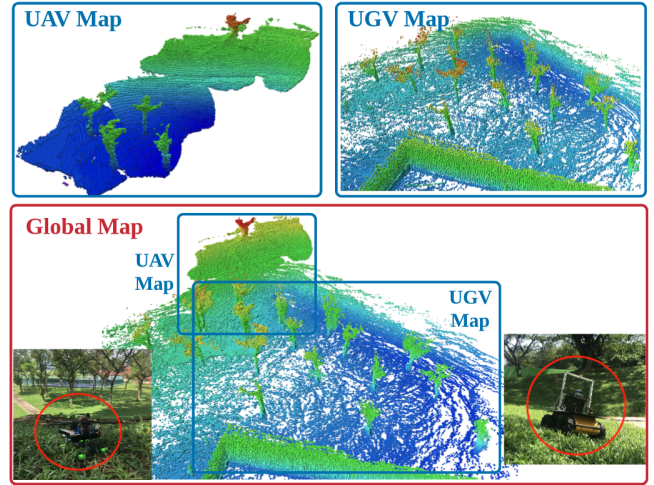


Fig. 5. The UAV-UGV collaborative mapping results in NTU Rainforest.

triggered by human and 6 times triggered by map information gain. The threshold to trigger UGV map fusion is set as 4000, because the equipped Lidar has 360 degree field of view. It can be observed that the information gain of UGV was increasing slowly before 140s, because it was moving slowly on rough terrain. For the UAV, the information gain trigger threshold is 500 given its limited sensing range and field of view. At around 50 seconds, the drone's information gain was increasing slowly because it encountered strong wind and was moving slowly to avoid tree trunks.

2) *UAV-UGV Fused Map*: Due to the limited mobility of UGV, the ramp was too steep for UGV to go through, therefor UGV turned to map the flat terrain, and UAV flew uphill and reconstructed the slope. It can be learned that in this unstructured environment, the UAV system we developed is able to localize accurately and generate an accurate local map. Fig. 5 shows the final fused map of combing partial maps. The algorithm combines the sparse UGV map and dense UAV map and generates an enhanced and more consistent global map.

C. Two-Level Building

In this environment, two robots started from a nearby place, the UGV could not go through the staircase and went through the slope, while the UAV directly flew over the staircase.

1) *Trigger Time Line*: In Fig. 6, the information gain witnesses a sharp growth in the beginning for both robots, because the robots just started the mapping task and obtained plenty environmental information. Then, UGV went through a narrow passage, therefore the information gain is increasing slowly and the 1st map fusion for UGV is triggered after 50 seconds. After UGV passed the passage and entered the open space, the map information is increased rapidly and 2nd map fusion is triggered in a short time interval. For the UAV, it hovered for a while and then started to fly over the staircase, therefore the information gain is also increasing slowly until the first map fusion triggered at around 60 seconds.

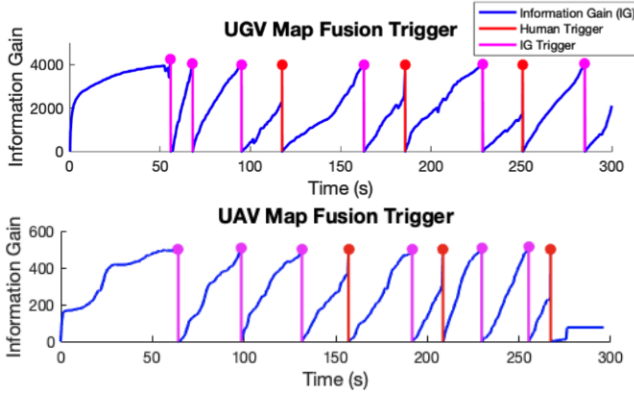


Fig. 6. The map fusion trigger slots in Two-Level Building.

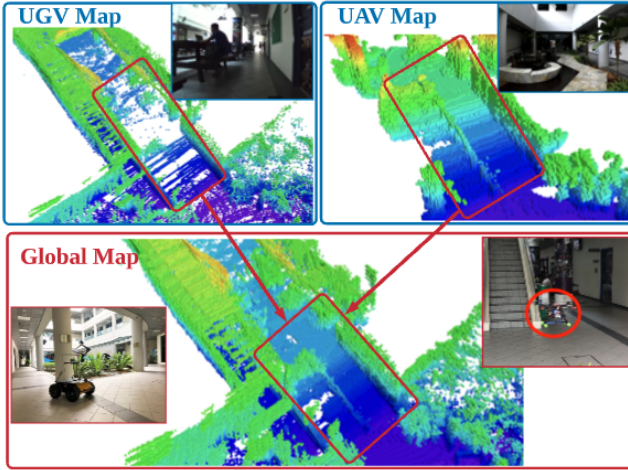


Fig. 7. The UAV-UGV collaborative mapping results in Two-Level Building.

2) *Fused map*: In Fig. 7, UGV failed to map the staircase and went through the narrow pass. UAV flew over the staircase and mapped the area. As the mission continues, the UAV fully mapped the staircase and affiliated garden, where the UGV missed that area. UGV map is also sparse due to sensor sparsity. By combining the local maps generated by UAV and UGV, a more detailed global consistent map is reconstructed. The combined map is comprehensive and dense, filled in the gaps and holes of UGV map.

D. Overall Performance Analysis

1) *Decision Cycle Frequency*: The decision frequency of single robot and collaborative robots is a key indicator to evaluate efficiency. The paper proposes a continuous-discrete

TABLE I
AVERAGE DECISION CYCLE (SECONDS)

Experiments	Single Robot		Collaborative Robots	
	UGV Mapping	UAV Mapping	UGV Fusion	UAV Fusion
NTU Rainforest	0.045	0.1042	43.71	33.53
Two-level Building	0.046	0.0867	30.87	29.69

TABLE II
THE COMPARISON OF AVERAGE ENTROPY IN ALL EXPERIMENTS. (THE BEST PERFORMANCE IS DENOTED IN BOLD.)

Experiments	Single Map		Global Map	
	UGV Map	UAV Map	Jessup [16]	Ours
NTU Rainforest	0.303	0.322	0.289	0.273
Two-level Building	0.278	0.293	0.269	0.257

model for decision-making: single robot continuous mapping and UAV-UGV discrete map fusion. Tab. I summarizes the average decision cycle on single robot and collaborative robots level. Single robot continuous mapping achieves real-time objectives such as localization and mapping. For UAV continuous mapping, the frequency is around 10Hz, while UGV is around 20Hz. The reason is UGV has stronger computation power, and executes at a higher frequency. Considering connectivity, bandwidth, and computational constraints, UAV-UGV discrete map fusion processes is performed from time to time. As can be observed in Fig. 4, 6, the frequency of UAV-UGV map fusion is usually greater than 20 seconds. In Tab. I, since the environment is highly unstructured, map fusion is performed much slower to perform in NTU Forest.

2) *Map Fusion Accuracy*: Table II summarizes the average entropy after performing map fusion in three experiments. In Table II, the value presents the average entropy of the probabilistic map in the different experiment. It can be observed that the average entropy of UAV map is higher than UGV map, because UAV is equipped stereo camera and has higher noise, therefore UAV map has higher uncertainty. After map fusion, we compared the average entropy between Jessup [16] and our algorithm. Jessup proposes a hard one-to-one fusion strategy to decrease entropy, but does not consider the multi-resolution problem. Our hybrid map fusion strategy gives the flexibility of fusing multiple voxels, the average entropy is further decreased compared to Jessup [16]. Experiments show that the proposed map fusion strategy is suitable for multi-resolution map fusion, and is able to decrease average entropy.

VI. CONCLUSIONS

This paper has presented an approach for collaborative UAV-UGV mapping in GNSS-denied environment. This paper has proposed a unified framework to formulate UAV-UGV collaborative mapping problem with a continuous-discrete model, as well as its realization in real robotic systems. The continuous space allows each robot to perform high frequency local map estimation, while the discrete space describes multi-resolution hybrid map fusion based on information gain trigger scheme. The results have shown that the proposed algorithm is able to perform collaborative mapping in GNSS-denied and unstructured environment. In summary, the proposed collaboration system provides a new perspective of perception and collaboration between heterogeneous robots, which is complementary to the individual robot perception and mapping.

REFERENCES

- [1] Y. Yue and D. Wang, *Collaborative Perception, Localization and Mapping for Autonomous Systems*. Springer, 2020.
- [2] Y. Rizk, M. Awad, and E. W. Tunstel, "Cooperative heterogeneous multi-robot systems: A survey," *ACM Computing Surveys (CSUR)*, vol. 52, no. 2, p. 29, 2019.
- [3] M. Krizmanic, B. Arbanas, T. Petrovic, F. Petric, and S. Bogdan, "Co-operative aerial-ground multi-robot system for automated construction tasks," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 798–805, 2020.
- [4] Q. Zhang, M. Wang, Y. Yue, and T. Liu, "Lcr-smm: Large convergence region semantic map matching through expectation maximization," *IEEE/ASME Transactions on Mechatronics*, 2021.
- [5] Y. Wang, M. Shan, Y. Yue, and D. Wang, "Autonomous target docking of nonholonomic mobile robots using relative pose measurements," *IEEE Transactions on Industrial Electronics*, pp. 1–11, 2020.
- [6] J. L. Paneque and et al, "Multi-sensor 6-dof localization for aerial robots in complex gnss-denied environments," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 1978–1984.
- [7] J. Park, P. Kim, Y. K. Cho, and J. Kang, "Framework for automated registration of uav and ugv point clouds using local features in images," *Automation in Construction*, vol. 98, pp. 175 – 182, 2019.
- [8] H. Surmann, N. Berninger, and R. Worst, "3d mapping for multi hybrid robot cooperation," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Sept 2017, pp. 626–633.
- [9] A. Gawel, R. Dubé, H. Surmann, J. Nieto, R. Siegwart, and C. Cadena, "3d registration of aerial and ground robots for disaster response: An evaluation of features, descriptors, and transformation estimation," in *2017 IEEE International Symposium on Safety, Security and Rescue Robotics (SSRR)*, Oct 2017, pp. 27–34.
- [10] Y. Tian, K. Liu, K. Ok, L. Tran, D. Allen, N. Roy, and J. P. How, "Search and rescue under the forest canopy using multiple uavs," *The International Journal of Robotics Research*, 2020.
- [11] A. Gawel, Y. Lin, T. Koutros, R. Siegwart, and C. Cadena, "Aerial-ground collaborative sensing: Third-person view for teleoperation," in *2018 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, Aug 2018, pp. 1–7.
- [12] C. Potena, R. Khanna, J. Nieto, R. Siegwart, D. Nardi, and A. Pretto, "Agricolmap: Aerial-ground collaborative 3d mapping for precision farming," *IEEE Robotics and Automation Letters*, vol. PP, pp. 1–1, 01 2019.
- [13] C. Forster, M. Pizzoli, and D. Scaramuzza, "Air-ground localization and map augmentation using monocular dense reconstruction," in *2013 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2013, pp. 3971–3978.
- [14] S. Saeedi, M. Trentini, M. Seto, and H. Li, "Multiple-Robot Simultaneous Localization and Mapping: A Review," *Journal of Field Robotics*, vol. 33, no. 1, pp. 3–46, 2016.
- [15] C. Brand, M. J. Schu, and H. Hirsch, "Submap matching for stereo-vision based indoor/outdoor SLAM," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, Sept 2015, pp. 5670–5677.
- [16] J. Jessup, S. N. Givigi, and A. Beaulieu, "Robust and efficient multirobot 3-d mapping merging with octree-based occupancy grids," *IEEE Systems Journal*, vol. 11, no. 3, pp. 1723–1732, 2017.
- [17] Y. Yue, C. Yang, Y. Wang, P. C. N. Senaratne, J. Zhang, M. Wen, and D. Wang, "A multilevel fusion system for multirobot 3-d mapping using heterogeneous sensors," *IEEE Systems Journal*, vol. 14, no. 1, pp. 1341–1352, 2019.
- [18] Y. Yue, C. Yang, J. Zhang, M. Wen, Z. Wu, H. Zhang, and D. Wang, "Day and night collaborative dynamic mapping in unstructured environment based on multimodal sensors," in *Robotics and Automation (ICRA), 2020 IEEE International Conference on*, May 2020.
- [19] T. I. Zohdi, "Multiple uavs for mapping: A review of basic modeling, simulation, and applications," *Annual Review of Environment and Resources*, vol. 43, pp. 523–543, 2018.
- [20] D. Scaramuzza, M. C. Achtelik, L. Doitsidis, F. Friedrich, and E. Kosmatopoulos, "Vision-Controlled Micro Flying Robots: From System Design to Autonomous Navigation and Mapping in GPS-Denied Environments," *IEEE Robotics Automation Magazine*, vol. 21, no. 3, pp. 26–40, Sept 2014.
- [21] Y. Lyu, Q. Pan, Y. Zhang, C. Zhao, H. Zhu, T. Tang, and L. Liu, "Simultaneously multi-uav mapping and control with visual servoing," in *2015 International Conference on Unmanned Aircraft Systems (ICUAS)*. IEEE, 2015, pp. 125–131.
- [22] M. Ani Hsieh and et al, "Adaptive teams of autonomous aerial and ground robots for situational awareness: Field reports," *Journal of Field Robotics*, vol. 24, pp. 991 – 1014, 11 2007.
- [23] S. Zhang and et al, "An autonomous air-ground cooperative field surveillance system with quadrotor uav and unmanned atv robots," in *2018 IEEE International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*, July 2018, pp. 1527–1532.
- [24] N. Michael, S. Shen, K. Mohta, Y. Mulgaonkar, V. Kumar, K. Nagatani et al., "Collaborative mapping of an earthquake-damaged building via ground and aerial robots," *Journal of Field Robotics*, vol. 29, no. 5, pp. 832–841, 2012.
- [25] H. Qin, Z. Meng, W. Meng, X. Chen, H. Sun, F. Lin, and M. H. Ang, "Autonomous exploration and mapping system using heterogeneous uavs and ugvs in gps-denied environments," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 2, pp. 1339–1350, 2019.
- [26] J. Zhang and S. Singh, "Loam: Lidar odometry and mapping in real-time," in *Robotics: Science and Systems Conference*, July 2014.
- [27] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardos, "Orb-slam: a versatile and accurate monocular slam system," *IEEE transactions on robotics*, vol. 31, no. 5, pp. 1147–1163, 2015.
- [28] A. Hornung, K. M. Wurm, M. Bennewitz, C. Stachniss, and W. Burgard, "OctoMap: An Efficient Probabilistic 3D Mapping Framework Based on Octrees," *Autonomous Robots*, 2013.
- [29] R. B. Rusu and S. Cousins, "3D is here: Point Cloud Library (PCL)," in *IEEE International Conference on Robotics and Automation (ICRA)*, Shanghai, China, May 9-13 2011.